

Final Reflection  
TCSS-498 Directed Reading (Raghavi Sakpal)  
Seth Campbell

This is a final reflection to recap my thoughts on what I've learned during this directed reading and over the course as a whole. I do want to say that this opportunity to dive deeper into a computer engineering-related interest has been tremendous. I feel I've learned so much in the 9 weeks since I started, and I'm ending my first directed reading with a deeper understanding of the basics of robotics but also with more questions. I am looking at future-directed readings, but before talking about that, I'd like to first cover the good, the bad, and the ugly of this directed reading.

Let's cover the ugly first. I started this directed reading with a general idea; I knew where I wanted to start and how I wanted to approach it. However, I got a reality check that my goal and aim were very ambitious even under ideal conditions. I wanted to use the ESP32 chip and structure because of other IoT projects I have slated for the future, and this came with a learning curve I had to deal with while doing the directed reading. This theme repeated itself several times throughout the project, and if I had backed up OR planned better, I could have had greater implementation of the material I had read (fully developed FK and IK). In the future, for other-directed readings, I want to do the following: Be familiar with the infrastructure before the direct reading; Minimize the 'ambition' of deliverables; plan for the unexpected.

The bad is singular in nature, but it's going to cover my biggest regret for the course. I wasn't able to deliver on the implementation I had initially set out to develop. I admit it was ambitious, given some pitfalls and troubles; however, I feel if I had caught myself sooner, I could have narrowed the scope a bit and focused more on developing the kinematics. I spent a lot of time tracking down faults that didn't matter to the end goal, and this ultimately prevented a better-developed kinematics implementation.

The good, I cannot overstate this, regardless of how developed my implementation was, I feel I got a lot out of this reading. Not just in an understanding of robotic motion and control but also in terms of communication protocols, ESP development, a deeper understanding of memory control, and a chance to dive into C++ again. Being able to explore a topic like this and to do so with a level of creative freedom and flexibility has been wonderful.

Looking forward, I have 2-3 planned directed readings that I hope to do during my undergrad. The next one will most likely be in the winter of next year. This will involve developing the machine learning integration and carrying out a hybrid approach of supervised leader/follower training and simulation. However, taking my lessons learned from this quarter. I will need to ensure I've completed the following prior to doing this: Transfer the leader/follower system to the larger arm and scale accordingly (small leader / larger follower); settle on AI vision implementation and integrate it; settle on a training environment for simulation. Luckily, what I've written will translate very well to the larger follower.

I've really appreciated Raghavi Sakpal for being my advisor for this and giving me the opportunity to explore my interest in robotics.